

SHREYASH AGRAWAL

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PROFILE

B.Tech Information Technology student at GLA University with strong full-stack, ML/DL, and LLM engineering skills. Experienced in building AI-powered applications using LangChain, LangGraph, RAG Pipelines, and Multi-Agent systems on top of a full-stack foundation. Passionate about solving real-world problems through scalable, production-ready software.

EDUCATION

B.Tech in Information Technology, GLA University

August 2024 – Present

CGPA: 7.5/10

KMPS School, Mathura

12th: 75% • 10th: 90%

SKILLS

Programming Languages: C, C++, Java, Python, JavaScript, TypeScript

Web Development: React.js, Node.js, Express.js, REST APIs, HTML, CSS, Tailwind CSS, Bootstrap

Backend & Databases: FastAPI, MongoDB, MySQL, Redis, Firebase, JWT Authentication

LLM Engineering: LangChain, LangGraph, RAG Pipelines, Multi-Agent Systems, FAISS, ChromaDB, HuggingFace Embeddings, Prompt Engineering, Groq API, Mistral AI, Streaming, Conversation Memory

AI/ML & Deep Learning: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, OpenCV, XGBoost, LightGBM, SHAP, Streamlit

Deep Learning Concepts: ANN, CNN, RNN, LSTM, Transfer Learning (ResNet50), NLP, TF-IDF, Cosine Similarity

Tools & DevOps: Git, GitHub, Docker, Docker Compose, nginx, AWS EC2, Postman, Google Earth Engine

WORK EXPERIENCE

Full Stack Developer Intern | Codec Technologies Pvt. Ltd.

May 2026 – Jun 2026

- Built **full-stack web applications** using **React.js, Node.js, Express.js, MongoDB** and **MySQL**, implementing **RESTful APIs** and database schemas with optimized **CRUD operations**.
- Collaborated with the team on UI design, business logic, and database integration using **Git & GitHub**, following **agile methodologies**.
- Debugged, tested, and optimized application reliability through end-to-end **deployment workflows**.

PROJECTS

CodeSmith AI — Autonomous Multi-Agent Software Engineering Platform

- Built an autonomous **multi-agent AI platform** using **LangGraph** with 11 specialized agents (PM, Architect, Database Designer, Backend, Frontend, Reviewer, Security, QA, BugFixer, TechWriter, DevOps) that generate production-ready full-stack applications from a single prompt, with **parallel backend + frontend execution**, **2 human-in-the-loop approval gates**, and an **iterative self-correction loop** (up to 3 cycles) driven by real compiler feedback.
- Implemented **real-time build validation** (npm build + py_compile), **WebSocket live streaming**, **LLM fallback routing** across Groq, Gemini, Mistral, Cerebras and NVIDIA, **Pydantic v2** structured outputs, SQLAlchemy job persistence, and IDE-style file explorer UI; deployed via **Docker Compose** with React + Vite frontend and FastAPI backend.

AI Code Reviewer — LangChain + RAG System

- Built a full-stack AI code review platform using **LangChain chains**, **Groq LLM**, and **Pydantic v2** for structured bug detection across 16+ languages with **streaming responses** and **conversation memory**; implemented a **RAG pipeline** for GitHub repo Q&A using **HuggingFace embeddings**, **FAISS vector store**, file prioritization, and source citations; deployed on **Streamlit Cloud**.

AQI Quality Prediction — ISRO BAH 2026 | Satellite + ML Platform

- Built a satellite-based AQI estimation platform for **ISRO's BAH 2026** problem statement using **Google Earth Engine (Sentinel-5P TROPOMI, MODIS AOD)** and **NASA FIRMS API**; trained an ensemble of **XGBoost, LightGBM, CatBoost, Random Forest** models ($R^2 > 0.85$) with **SHAP explainability**, **Z-score HCHO hotspot detection**, and **7-day autoregressive forecasting** with bootstrap confidence intervals; deployed on **Streamlit Cloud**.

FakeShield AI — Deepfake Detection System

- Built a **CNN-based ResNet50** transfer learning model for image classification and **TimeDistributed CNN + LSTM** for video, capturing spatio-temporal inconsistencies across 20-frame sequences; deployed via **FastAPI + React** frontend containerized with **Docker + nginx** and **Docker Compose**, images pushed to **Docker Hub**.

ADDITIONAL INFORMATION

- HTML, CSS, JS & Bootstrap** — Infosys Springboard | **AI & Cybersecurity Awareness** — TCS iON, May 2026 | **Generative AI Essentials** — TCS iON, Jun 2026
- Data Science & Analytics** — HP LIFE, Jun 2026 | **Trust and Security with Google Cloud** — Simplilearn, Jun 2026
- Solved 250+ DSA problems**, improving algorithmic thinking and coding efficiency. | **Languages:** English, Hindi.